Project Proposal: Agentic Data Fusion for Insurance Knowledge Graphs

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1 Problem Statement and Motivation

The insurance industry relies on decision-making processes that span highly heterogeneous data sources. Critical business logic is often fragmented across unstructured policy documents (PDFs), visual diagrams (flowcharts), and structured relational databases (claims tables). This fragmentation creates both technical and business challenges: data silos prevent unified reasoning, resulting in slow time-to-insight and inconsistencies in coverage decisions.

In practice, a commercial property claim involving water damage may require the adjuster to consult a database for customer and policy information, review policy-form language to determine whether the loss is consistent with covered perils or exclusions, and follow a decision tree to compute limits and deductibles. Because these sources cannot be jointly analyzed, determining whether the claim is covered often requires extensive manual reconciliation. This project aims to address this challenge by developing an automated, end-to-end pipeline that fuses heterogeneous insurance data into a unified **Knowledge Graph (KG)**, enabling explainable cross-source analytics and consistent decision support.

2 Background and Existing Solutions

Most retrieval systems rely on **Vector RAG**, which achieves strong results for semantic similarity but fails at multi-hop reasoning and global summarization [Edge et al., 2024]. RAG alone cannot reliably connect a clause in a policy-form PDF to a structured claim record or derive an explanation that depends on multiple linked facts.

Traditional Knowledge Graph construction pipelines are rigid and rely heavily on handcrafted ETL logic. Recent work argues that autonomous LLM agents can automate extraction, disambiguation, and enrichment [Pan et al., 2024]. The 2025 **KARMA** framework [Lu et al., 2025] demonstrates that multi-agent LLM systems can iteratively debate, refine, and validate Knowledge Graphs, substantially reducing duplication and improving consistency. Motivated by KARMA and multi-step reasoning frameworks such as Graph of Thoughts [Besta et al., 2024], this project adapts these ideas to a multimodal insurance context.

3 Initial Approach / Baseline (Phase 1)

Before Spring Break, we will develop a **simple extraction baseline** to establish a lower bound on performance.

Baseline Pipeline

- **Ingestion:** PDFs are loaded using `PyPDFLoader` and chunked into 512-token segments with 50-token overlap.
- **Extraction:** A locally run **Llama 3.1 8B** model extracts triplets under a minimal schema (e.g., `Policy`, `Claim`, `Person`) without grammar constraints or validation.
- **Storage:** Extracted triplets are inserted into Neo4j.

Expected Limitations

This baseline ignores inter-document reasoning and thus produces (i) high duplication (e.g., variants of names), (ii) weak linkage between structured and unstructured sources, (iii) no extraction from flowcharts, and (iv) no mechanism for conflict detection or validation. These issues will be quantified using precision/recall for entity matching on a manually annotated sample.

4 Proposed Innovation: Multi-Agent "KARMA-Inspired" Pipeline

The main contribution of this project is a **multi-agent extraction, fusion, and validation pipeline** implemented in LangGraph and architecturally inspired by the 2025 KARMA framework [Lu et al., 2025] and Graph of Thoughts [Besta et al., 2024].

Model Strategy (Resource-Constrained)

Due to limited hardware resources, we will prioritize **local** or lightweight models:

- **Primary models:** Distilled 7B–8B variants of **DeepSeek-V3** and **DeepSeek-R1**, chosen for strong logical consistency in KG tasks.
- **Baselines:** Llama 3.1 8B or smaller variants.
- **Optional upper bound:** A small number of calls to a hosted DeepSeek-V3 model.

Four-Zone Multi-Agent Architecture

- **Zone 1: Multimodal Ingestion Agent.** Routes PDFs, tables (CSV), and images to the appropriate processing tools. For flowcharts, we integrate **Qwen2.5-VL**; if local hardware cannot support it, we fall back to OCR + text models.
- **Zone 2: Structured Extraction Agent.** Converts parsed text into ontology-aligned JSON using grammar-constrained prompts. The ontology includes entities such as `Policy`, `Coverage`, `Exclusion`, `Risk`, and `Claim`.
- **Zone 3: Validation and "KARMA Loop" Agent.** Identifies contradictory or duplicate entities (e.g., variants of names/addresses or conflicting dates) and resolves them via multi-agent debate and rule-based checks before inserting or merging nodes in Neo4j.
- **Zone 4: Storage and Query.** The validated KG is stored in Neo4j and queried via Cypher; a lightweight Streamlit UI supports interactive inspection and question answering.

Figure 1 summarizes the end-to-end pipeline and highlights the KARMA-style validation loop that reduces duplication and improves consistency before writing to the graph database.

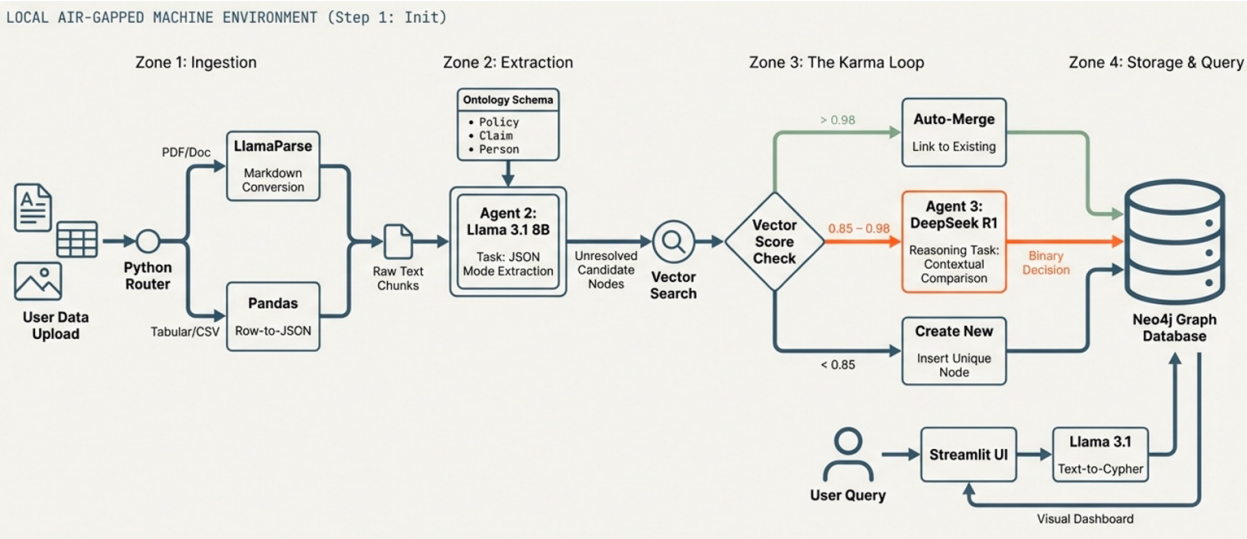

Figure 1: Air-gapped, multi-agent pipeline for insurance KG construction. Zone 1 ingests PDFs/tables via a router; Zone 2 performs schema-constrained extraction; Zone 3 applies the KARMA-style validation loop (vector retrieval + LLM verification) to merge or create entities; Zone 4 stores the KG in Neo4j and supports query through a lightweight UI.

5 Datasets

We will evaluate our system on two complementary tracks that together cover both same-domain fusion and cross-domain portability.

Track A (NFIP flood/property, end-to-end fusion). We use structured NFIP transaction data from OpenFEMA, including **FIMA NFIP Redacted Policies - v2** [fem, b] and **FIMA NFIP Redacted Claims - v2** [fem, a]. To supply true unstructured policy-form clause text (definitions, coverages, exclusions), we include a curated set of official **Standard Flood Insurance Policy (SFIP)** PDFs, such as the General Property Form [fem, c].

Track B (auto, portability study). To test generalization to a second line of business, we use the Kaggle **Auto Insurance Claims** dataset [kag, a] and pair it with a Kaggle **Vehicle Insurance Customer/Policy** dataset containing structured policy-like attributes [kag, b].

Fusion Objective. In Track A, link NFIP claims to the corresponding policy entities and extracted SFIP clauses (e.g., mapping a claim to relevant coverages/exclusions and explaining the decision path). In Track B, link auto claims to policy/customer entities and compare extraction and fusion behavior across domains.

6 Evaluation Method and Metrics

We evaluate both KG construction quality and downstream utility, comparing the baseline pipeline to the KARMA-inspired multi-agent pipeline.

- **Entity Resolution Accuracy.** We manually annotate a "Golden Subgraph" of 50 cases (policy/claim records plus associated policy-form clauses). We compute precision/recall/F1 for entity merging and identity linking (e.g., customer/policy/claim and clause entities).
- **Relation Accuracy.** On the same gold subset, we annotate key edges (e.g., `Claim`→`Policy`, `Policy`→`Exclusion/Coverage`, `Exclusion`→`Risk`) and report edge-level precision/recall/F1.

- **KG Validity via Constraints.** We define a small set of schema constraints (e.g., every claim links to a policy; exclusions belong to a policy) and report constraint violation rates as a measure of global consistency.
- **Business Query Accuracy.** We define 20 realistic insurance questions (split across Track A and Track B) and score answer correctness and evidence quality. We use an LLM judge not used in the construction pipeline with a fixed rubric, and manually spot-check a subset for reliability.
- **Explainability.** We visualize representative query paths in Neo4j Bloom to demonstrate how extracted clauses and structured attributes jointly support claim-related conclusions.

7 Deliverables

- **Mid-Term:** Baseline pipeline and initial Neo4j graph populated with a sampled subset of Track A data.
- **Final Codebase:** LangGraph multi-agent implementation of the four-zone pipeline.
- **Unified Knowledge Graph:** Exported Neo4j database containing fused multi-source data (Track A) and portability experiments (Track B).
- **Demo Application:** Streamlit interface for document upload and KG-driven query answering.
- **Final Report:** Ontology, agent architecture, evaluation metrics, and findings.

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